Optimization In AI

Class 05: Momentum (Deterministic Setting)

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Today's Topics

01 From GD To Momentum

02 Heavy Ball On Quadratics

03 Nesterov Acceleration

04 Lower Bounds

PART 01

From GD To Momentum

Recall: Gradient Descent

$$x_{t+1} = x_t - \eta \nabla f(x_t).$$

Smooth Convex Guarantee

For convex, L-smooth f with a minimizer $x^\star$, let $\eta = 1/L$.

$$f(x_T) - f^\star \leq \frac{L\|x_0 - x^\star\|^2}{2T}, \qquad T \geq 1.$$

Heavy-Ball Momentum

Algorithm

Choose $\eta_t > 0$, $0 \leq \beta_t < 1$. Initialize $x_{-1} = x_0$. For $t = 0, 1, \ldots$, set

$$x_{t+1} = x_t - \eta_t \nabla f(x_t) + \beta_t (x_t - x_{t-1}).$$

Heavy Ball: Gradient Buffer

Equivalent Form

Choose constant $\eta > 0$, $0 \leq \beta < 1$. Initialize $x_{-1} = x_0$ and $b_0 = 0$. For $t = 0, 1, \ldots,$ set

$$b_{t+1} = \beta b_t + \nabla f(x_t),$$

$$x_{t+1} = x_t - \eta b_{t+1}.$$

Heavy Ball: Gradient Averaging

Equivalent Form

Choose constant $\eta > 0$, $0 \leq \beta < 1$. Initialize $x_{-1} = x_0$ and $m_0 = 0$. For $t = 0, 1, \ldots$, set

$$m_{t+1} = \beta m_t + (1 - \beta)\nabla f(x_t),$$

$$x_{t+1} = x_t - \frac{\eta}{1 - \beta} m_{t+1}.$$

Heavy Ball: Extrapolation

Equivalent Form

Choose constant $\eta > 0$, $0 \leq \beta < 1$. Initialize $x_{-1} = x_0$ and $y_0 = x_0$. For $t = 0, 1, \ldots$, set

$$y_t = x_t + \beta(x_t - x_{t-1}),$$

$$x_{t+1} = y_t - \eta \nabla f(x_t).$$

Heavy Ball: Primal Averaging

Equivalent Form

Choose constant $\eta > 0$, $0 \leq \beta < 1$. Initialize $x_{-1} = x_0$ and $q_0 = x_0$. For $t = 0, 1, \ldots,$ set

$$q_{t+1} = q_t - \frac{\eta}{1-\beta}\nabla f(x_t),$$

$$x_{t+1} = \beta x_t + (1-\beta) q_{t+1}.$$

Heavy Ball: Gradient History

Equivalent Form

Choose constant $\eta > 0$, $0 \leq \beta < 1$. Initialize $x_{-1} = x_0$. For $t = 0, 1, \ldots,$

$$x_{t+1} = x_t - \eta \sum_{s=0}^{t} \beta^{t-s} \nabla f(x_s).$$

Heavy Ball: The Final Iterate

Equivalent Form

Choose constant $\eta > 0$, $0 \leq \beta < 1$. Initialize $x_{-1} = x_0$. For every integer $T \geq 1$,

$$x_T = x_0 - \frac{\eta}{1-\beta} \sum_{s=0}^{T-1} \left(1 - \beta^{T-s}\right) \nabla f(x_s).$$

Nesterov Accelerated Gradient

Algorithm

Choose $\eta_t > 0$, $0 \leq \beta_t < 1$. Initialize $x_{-1} = x_0$. For $t = 0, 1, \ldots$, set

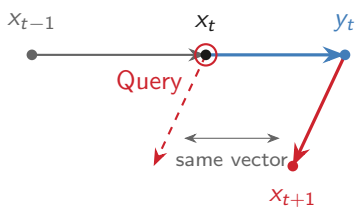
$$y_t = x_t + \beta_t(x_t - x_{t-1}),$$

$$x_{t+1} = y_t - \eta_t \nabla f(y_t).$$

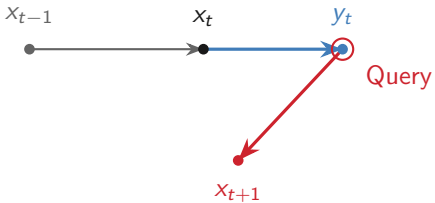
Heavy Ball And Nesterov: Two Updates

$$y_t = x_t + \beta_t(x_t - x_{t-1}).$$

Heavy Ball



Nesterov



$$x_{t+1} = y_t - \eta_t \nabla f(x_t).$$

$$x_{t+1} = y_t - \eta_t \nabla f(y_t).$$

Extrapolation Gradient Step

PART 02

Heavy Ball On Quadratics

Convex Quadratics And Zero Curvature

Assumption (Smooth Convex Quadratic Model)

$$f(x) = f^\star + \frac{1}{2}(x - x^\star)^\top Q(x - x^\star), \qquad 0 \preceq Q = Q^\top \preceq LI, \quad L > 0.$$

$$e_t = x_t - x^\star, \qquad \|e_0\| \le R, \qquad \Delta_t = f(x_t) - f^\star = \tfrac{1}{2} e_t^\top Q e_t.$$

Recall: Spectral Polynomials And Chebyshev

Quadratic GD

$$e_T = p_T(Q)e_0, \qquad p_T(\lambda) = \prod_{t=0}^{T-1}(1-\eta_t\lambda) \quad \text{for GD.}$$

$$Qu = \lambda u \quad \Longrightarrow \quad \langle u, e_T\rangle = p_T(\lambda)\langle u, e_0\rangle.$$

First-Kind Chebyshev Polynomials

$$C_0(z) = 1, \quad C_1(z) = z, \quad C_{t+1}(z) = 2zC_t(z) - C_{t-1}(z).$$

$$C_t(\cos\theta) = \cos(t\theta).$$

Heavy Ball: The Quadratic Schedule

Algorithm

Initialize $x_{-1} = x_0$. For $t = 0, 1, \ldots$, set

$$\beta_t = \frac{t}{t+2}, \qquad \eta_t = \frac{2(t+1)}{L(t+2)}$$

and update

$$x_{t+1} = x_t - \eta_t \nabla f(x_t) + \beta_t (x_t - x_{t-1}).$$

Convergence On Convex Quadratics

Theorem (Heavy Ball)

Let f be the quadratic model with $0 \preceq Q \preceq LI$, $L > 0$. Let $x^\star$ be a minimizer with $\|x_0 - x^\star\| \leq R$. Run HB with the quadratic schedule, initialized at $x_{-1} = x_0$. Then, for every integer $T \geq 0$,

$$f(x_T) - f^\star \leq \frac{L\|x_0 - x^\star\|^2}{2(T+1)^2} \leq \frac{LR^2}{2(T+1)^2}.$$

HB Proof: Rescale The Error

Quadratic Gradient $\nabla f(x_t) = Qe_t$

$$(t+2)e_{t+1} = 2(t+1)(I - Q/L)e_t - te_{t-1}.$$

Scaled State

$$B = I - Q/L, \qquad w_t = (t+1)e_t, \qquad w_{-1} = 0.$$

$$\boxed{w_{t+1} = 2Bw_t - w_{t-1}}, \qquad w_0 = e_0.$$

The HB Recurrence Uses Second-Kind Chebyshev

Second-Kind Chebyshev Polynomials

$$U_0(z) = 1, \quad U_1(z) = 2z, \quad U_{t+1}(z) = 2zU_t(z) - U_{t-1}(z).$$

$$w_0 = e_0, \quad w_1 = 2Be_0 \quad \Longrightarrow \quad w_t = U_t(B)e_0.$$

$$\boxed{e_t = \frac{U_t(I - Q/L)}{t+1} e_0}.$$

Zero Curvature $U_0(1) = 1,\ U_1(1) = 2,\ U_t(1) = t + 1$

$$Qu = 0 \quad \Longrightarrow \quad \langle u, e_t \rangle = \langle u, e_0 \rangle.$$

HB Proof: An Energy For The Scaled State

Definition (Quadratic Energy)

$$E_t = \|w_{t-1} - Bw_t\|^2 + w_t^\top (I - B^2) w_t.$$

Conservation And Gap Bound

$$E_{t+1} = E_t,$$

$$E_t \geq \frac{2(t+1)^2}{L} \Delta_t.$$

HB Proof: The Energy Is Conserved

Expand The Definition: $B = B^\top$

$$E_t = \|w_t\|^2 + \|w_{t-1}\|^2 - 2\langle w_{t-1}, Bw_t\rangle .$$

Use $w_{t+1} + w_{t-1} = 2Bw_t$

$$\begin{aligned} E_{t+1} - E_t &= \|w_{t+1}\|^2 - \|w_{t-1}\|^2 \\ &\quad - 2\langle Bw_t, w_{t+1} - w_{t-1}\rangle \\ &= \langle w_{t+1} - w_{t-1}, w_{t+1} + w_{t-1} - 2Bw_t\rangle \\ &= 0. \end{aligned}$$

HB Proof: The Energy Controls The Objective

Use $0 \preceq Q \preceq LI$

$$I - B^2 = \frac{2Q}{L} - \frac{Q^2}{L^2} \succeq \frac{Q}{L}, \qquad \lambda^2 \leq L\lambda \quad (0 \leq \lambda \leq L).$$

$$\boxed{E_t \geq \frac{1}{L} w_t^\top Q w_t = \frac{2(t+1)^2}{L} \Delta_t \geq 0}.$$

HB Proof: Recover The Accelerated Rate

Initialization $w_{-1} = 0, \quad w_0 = x_0 - x^\star$

$$\begin{aligned} E_0 &= \|Bw_0\|^2 + w_0^\top (I - B^2) w_0 \\ &= \|w_0\|^2 = \|x_0 - x^\star\|^2 \le R^2. \end{aligned}$$

Objective Gap Bound

$$\frac{2(T+1)^2}{L} \Delta_T \le E_T = E_0 \le R^2.$$

$$\boxed{\Delta_T \le \frac{LR^2}{2(T+1)^2}}.$$

Heavy Ball: Strongly Convex Quadratics

Theorem (Heavy Ball)

Let $f(x) = f^\star + \frac{1}{2}(x - x^\star)^\top Q(x - x^\star)$ with $\mu I \preceq Q = Q^\top \preceq LI$, $0 < \mu \leq L$, and $\|x_0 - x^\star\| \leq R$. Set $\kappa = L/\mu$ and initialize $x_{-1} = x_0$. Run heavy ball with constant parameters

$$\eta = \frac{4}{(\sqrt{L} + \sqrt{\mu})^2}, \qquad \beta = \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}\right)^2.$$

Then, for every integer $T \geq 1$,

$$\|x_T - x^\star\| \leq \left[1 + (1 + \rho)T\right]\rho^T R, \qquad \rho = \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}.$$

What Happens In A Zero-Curvature Direction?

For $Qu = 0$, let $a_t = \langle u, x_t - x^\star \rangle$.

$$a_{t+1} = a_t + \beta_t(a_t - a_{t-1}), \qquad a_{-1} = a_0.$$

$$a_t = a_0 \quad \text{for every } t \geq 0.$$

Objective Gap zero contribution from this direction.

Distance To A Chosen Minimizer need not vanish.

Simulation: A Singular Quadratic

Example (Smooth Convex, Not Strongly Convex)

$$Q = \mathrm{diag}(0, 1, 10^4), \qquad f(x) = \tfrac{1}{2} x^\top Q x, \qquad L = 10^4.$$

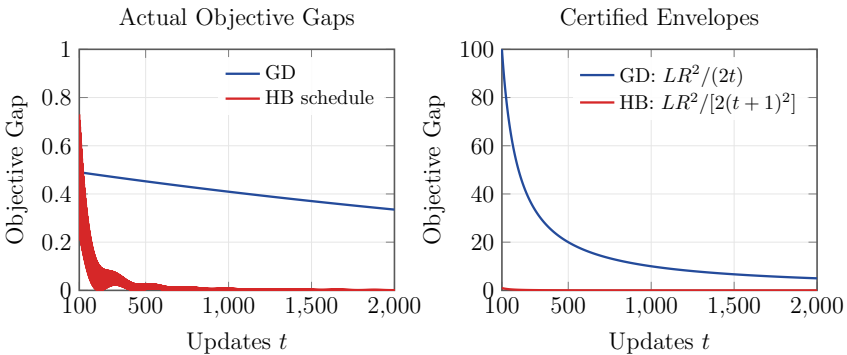
$$x_{-1} = x_0 = (1, 1, 1)^\top, \qquad x^\star = (1, 0, 0)^\top, \qquad R^2 = 2.$$

$$\text{GD: } \eta = 1/L; \qquad \text{HB: } \eta_t = \frac{2(t+1)}{L(t+2)}, \quad \beta_t = \frac{t}{t+2}.$$

$T = 2000$ exact-gradient updates.

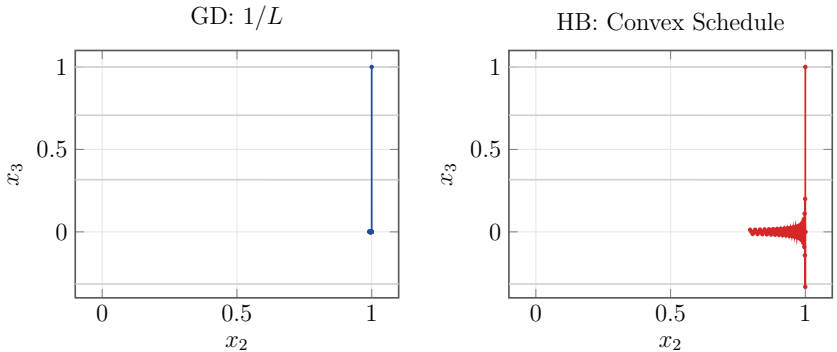
Simulation: GD Versus Scheduled Heavy Ball

$$Q = \mathrm{diag}(0, 1, 10^4), \quad x_0 = (1, 1, 1)^\top$$

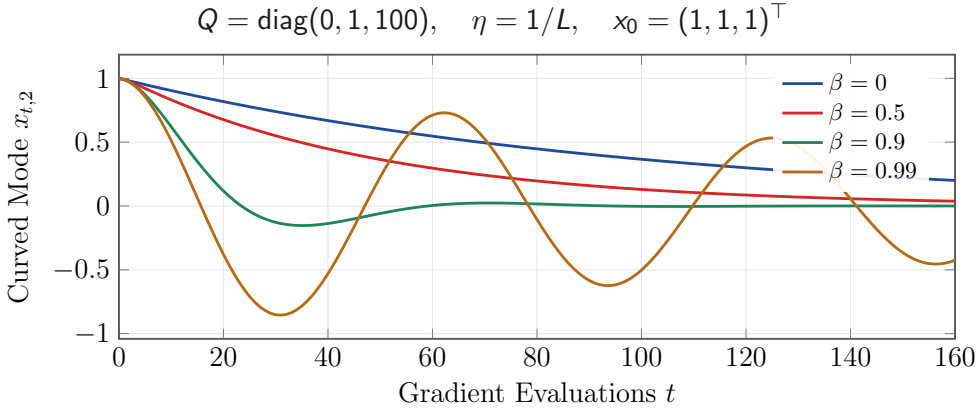


Simulation: Motion In Curved Directions

First 80 updates, projected onto (x_2, x_3).



Simulation: Fixed Momentum Values



What Did The Quadratic Proof Use?

Quadratic Gradient

$$\nabla f(x_t) = Q(x_t - x^\star) \quad \Longrightarrow \quad B = I - Q/L.$$

Conserved Energy

$$E_t = \|w_t\|^2 + \|w_{t-1}\|^2 - 2\langle w_t, Bw_{t-1}\rangle .$$

PART 03

Nesterov Acceleration

Nesterov: The Convex Schedule

Algorithm

Initialize $y_0 = x_0$, $s_1 = 1$ and

$$x_1 = x_0 - \frac{1}{L}\nabla f(x_0).$$

For $t = 1, 2, \ldots$, compute in order

$$s_{t+1} = \frac{1 + \sqrt{1 + 4s_t^2}}{2}, \qquad \beta_t = \frac{s_t - 1}{s_{t+1}},$$

$$y_t = x_t + \beta_t(x_t - x_{t-1}),$$

$$x_{t+1} = y_t - \frac{1}{L}\nabla f(y_t).$$

Acceleration On Smooth Convex Functions

Theorem (Nesterov)

Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex and L-smooth, $L > 0$. Let $x^\star \in \arg\min f$ and assume $\|x_0 - x^\star\| \le R$. Run NAG with the convex schedule and exact gradients. Then, for every integer $T \ge 1$,

$$f(x_T) - f^\star \le \frac{2L\|x_0 - x^\star\|^2}{(T+1)^2} \le \frac{2LR^2}{(T+1)^2}.$$

The Extrapolated State Hidden In NAG

NAG Extrapolation $t \geq 1$

$$y_t = x_t + \frac{s_t - 1}{s_{t+1}}(x_t - x_{t-1}).$$

Definition (Extrapolated State)

$$\begin{aligned} z_t &:= x_t + (s_t - 1)(x_t - x_{t-1}) \\ &= s_t x_t - (s_t - 1)x_{t-1}, \qquad z_1 = x_1. \end{aligned}$$

Substitute The Displacement

$$\boxed{y_t = x_t + \frac{z_t - x_t}{s_{t+1}} = \left(1 - \frac{1}{s_{t+1}}\right) x_t + \frac{1}{s_{t+1}} z_t.}$$

Two Updates From One Gradient

$g_t := \nabla f(y_t), \quad y_0 = x_0; \qquad$ below, $t \geq 1$.

Substitute $x_{t+1} = y_t - g_t/L$

$$\begin{aligned} z_{t+1} &= s_{t+1}x_{t+1} - (s_{t+1} - 1)x_t \\ &= \underbrace{s_{t+1}y_t - (s_{t+1} - 1)x_t}_{z_t} - \frac{s_{t+1}}{L}g_t. \end{aligned}$$

Equivalent NAG Updates

$$\begin{aligned} y_t &= \left(1 - \frac{1}{s_{t+1}}\right) x_t + \frac{1}{s_{t+1}} z_t, \\ x_{t+1} &= y_t - \frac{1}{L} g_t, \\ z_{t+1} &= z_t - \frac{s_{t+1}}{L} g_t. \end{aligned}$$

Proof: Smoothness And Weighted Convexity

$\Delta_t := f(x_t) - f^\star, \qquad s := s_{t+1} \geq 1, \quad t \geq 1.$

Smoothness: $x_{t+1} = y_t - g_t/L$

$$f(x_{t+1}) \leq f(y_t) - \frac{1}{2L}\|g_t\|^2.$$

Convexity At The Query y_t

$$f(x_t) \geq f(y_t) + \langle g_t, x_t - y_t \rangle,$$
$$f^\star \geq f(y_t) + \langle g_t, x^\star - y_t \rangle.$$

Weights $1 - 1/s$ **And** $1/s$**;** $\quad y_t = (1 - 1/s)x_t + z_t/s$

$$\left(1 - \frac{1}{s}\right) f(x_t) + \frac{1}{s} f^\star \geq f(y_t) - \frac{1}{s} \langle g_t, z_t - x^\star \rangle.$$

$$\boxed{\Delta_{t+1} \leq \left(1 - \frac{1}{s}\right) \Delta_t + \frac{1}{s} \langle g_t, z_t - x^\star \rangle - \frac{1}{2L}\|g_t\|^2.}$$

Proof: The Auxiliary Distance Difference

$$s = s_{t+1}, \qquad z_{t+1} = z_t - \frac{s}{L} g_t.$$

Expand The Squared Distance

$$\|z_{t+1} - x^\star\|^2 = \|z_t - x^\star\|^2 - \frac{2s}{L} \langle g_t, z_t - x^\star \rangle + \frac{s^2}{L^2} \|g_t\|^2.$$

$$s \langle g_t, z_t - x^\star \rangle - \frac{s^2}{2L} \|g_t\|^2 = \frac{L}{2} \left(\|z_t - x^\star\|^2 - \|z_{t+1} - x^\star\|^2 \right).$$

Multiply The One-Step Gap Bound By s^2

$$s^2 \Delta_{t+1} \le (s^2 - s)\Delta_t + s \langle g_t, z_t - x^\star \rangle - \frac{s^2}{2L} \|g_t\|^2.$$

$$\boxed{s^2 \Delta_{t+1} + \frac{L}{2} \|z_{t+1} - x^\star\|^2 \le (s^2 - s)\Delta_t + \frac{L}{2} \|z_t - x^\star\|^2.}$$

Proof: The Accelerated Potential

The NAG Schedule

$$2s_{t+1} - 1 = \sqrt{1 + 4s_t^2} \quad \Longrightarrow \quad \boxed{s_{t+1}^2 - s_{t+1} = s_t^2.}$$

Definition (Accelerated Potential)

$$\Phi_t = s_t^2 \Delta_t + \frac{L}{2}\|z_t - x^\star\|^2, \qquad t \geq 1.$$

Substitute Into The Weighted Gap–Distance Bound

$$\boxed{\Phi_{t+1} \leq \Phi_t \qquad (t \geq 1).}$$

	Gap Weight	Distance State
GD	t	x_t
NAG	s_t^2	z_t

Proof: Initialize The Potential

First Gradient Step

$$y_0 = x_0, \qquad g_0 = \nabla f(x_0), \qquad x_1 = x_0 - \frac{1}{L} g_0, \qquad s_1 = 1, \quad z_1 = x_1.$$

Smoothness And Convexity At x_0

$$\begin{aligned}
\Delta_1 &\leq f(x_0) - f^\star - \frac{1}{2L}\|g_0\|^2 \\
&\leq \langle g_0, x_0 - x^\star \rangle - \frac{1}{2L}\|g_0\|^2 \\
&= \frac{L}{2}\left(\|x_0 - x^\star\|^2 - \|x_1 - x^\star\|^2\right).
\end{aligned}$$

Use $s_1 = 1$ And $z_1 = x_1$

$$\boxed{\Phi_1 = \Delta_1 + \frac{L}{2}\|x_1 - x^\star\|^2 \leq \frac{L}{2}\|x_0 - x^\star\|^2.}$$

Proof: Quadratic Weight Growth Gives The Rate

Growth Of The Existing Schedule

$$s_{t+1} = \frac{1+\sqrt{1+4s_t^2}}{2} \geq s_t + \frac{1}{2}, \qquad s_1 = 1.$$

$$s_T \geq 1 + \frac{T-1}{2} = \frac{T+1}{2} \qquad (T \geq 1).$$

Nonnegative Distance; Potential Decrease; Initialization

$$s_T^2 \Delta_T \leq \Phi_T \leq \Phi_1 \leq \frac{L}{2}\|x_0 - x^\star\|^2.$$

$$\Delta_T \leq \frac{L\|x_0 - x^\star\|^2}{2s_T^2} \leq \boxed{\frac{2L\|x_0 - x^\star\|^2}{(T+1)^2} \leq \frac{2LR^2}{(T+1)^2}.}$$

NAG: Strongly Convex Functions

Theorem (Nesterov)

Let $f : \mathbb{R}^d \to \mathbb{R}$ be L-smooth and μ-strongly convex, $0 < \mu \leq L$, with minimizer $x^\star$. Set $\kappa = L/\mu$, initialize $x_{-1} = x_0$, and run NAG with constant parameters

$$\eta = \frac{1}{L}, \qquad \beta = \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}.$$

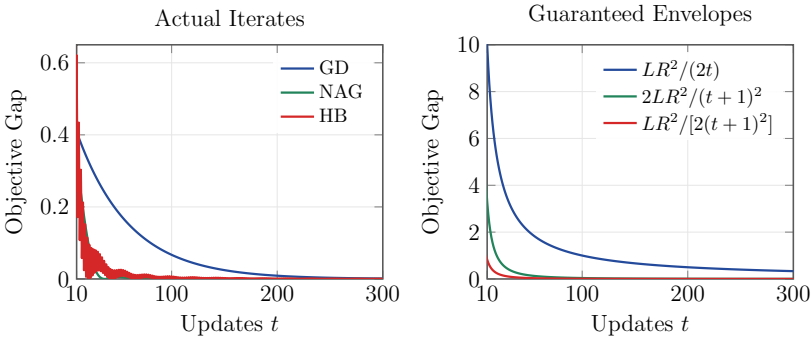
Then, for every integer $T \geq 1$,

$$f(x_T) - f^\star \leq \left(1 - \frac{1}{\sqrt{\kappa}}\right)^T \left[f(x_0) - f^\star + \frac{\mu}{2}\|x_0 - x^\star\|^2\right].$$

Simulation: Bounds And Actual Curves

$$Q = \mathrm{diag}(0, 1, 100), \quad x_0 = (1, 1, 1)^\top, \quad x^\star = (1, 0, 0)^\top$$

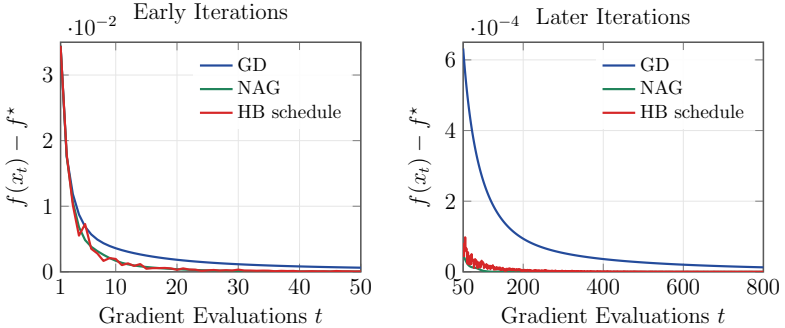
$R^2 = 2$; GD: $\eta = 1/L$; NAG/HB: time-varying convex schedules.



Simulation: A Nonquadratic Objective

$$f(x) = \frac{1}{64}\sum_{i=1}^{64} \log(1 + e^{-b_i a_i^\top x}), \quad x \in \mathbb{R}^8$$

Paired labels ± 1 for each feature row: $x^\star = 0$, $f^\star = \log 2$.



$A = [a_i^\top]_{i=1}^n$, $L = \|A\|_2^2/(4n)$, $x_0 = \mathbf{1}$; GD/NAG: $\eta = 1/L$; HB: quadratic schedule (no general guarantee).

What The Proof Actually Guarantees

	Heavy Ball	**Nesterov**
Function class	Smooth convex quadratic	Any smooth convex f
Certificate	Conserved energy	Decreasing potential
Last objective gap	$LR^2/[2(T+1)^2]$	$2LR^2/(T+1)^2$

PART 04

Lower Bounds

General First-Order Methods

Definition (Linear-Span Model)

Starting from x_0, make T exact gradient queries at $y_0, \dots, y_{T-1}$. Let

$$\mathcal{K}_0 = \{0\}, \qquad \mathcal{K}_t = \operatorname{span}\{\nabla f(y_0), \dots, \nabla f(y_{t-1})\}.$$

The queries and returned point satisfy

$$y_t \in x_0 + \mathcal{K}_t \quad (0 \le t < T), \qquad x_T \in x_0 + \mathcal{K}_T.$$

Coefficients may depend on all previous oracle responses.

A Lower Bound For First-Order Methods

Theorem (Nemirovski–Yudin)

For every integer $T \geq 1$ and $L, R > 0$, set $d = 2T + 1$. For any method in the preceding linear-span model initialized at $x_0 = 0$, there exists a convex, L-smooth quadratic $f : \mathbb{R}^d \to \mathbb{R}$ with a minimizer $x^\star$ satisfying $\|x^\star\| = R$ such that, after T gradient queries,

$$\boxed{f(x_T) - f^\star \geq \frac{3LR^2}{32(T+1)^2}.}$$

Proof: Construct A Quadratic Chain

Let $d = 2T + 1$, $a > 0$, and $e_1 = (1, 0, \ldots, 0)^\top \in \mathbb{R}^d$.

$$A_d = \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix} \in \mathbb{R}^{d \times d}.$$

$$f(x) = \frac{L}{8}\left(x^\top A_d x - 2a\, e_1^\top x\right), \qquad x_0 = 0.$$

Proof: Convexity And Smoothness

Quadratic Form

$$v^\top A_d v = v_1^2 + \sum_{i=1}^{d-1} (v_i - v_{i+1})^2 + v_d^2$$

$$\leq v_1^2 + 2\sum_{i=1}^{d-1} (v_i^2 + v_{i+1}^2) + v_d^2 \leq 4\|v\|^2.$$

The sum of squares is zero only when $v = 0$.

Hessian

$$0 \prec A_d \preceq 4I \qquad \Longrightarrow \qquad 0 \prec \nabla^2 f(x) = \frac{L}{4} A_d \preceq LI.$$

Proof: One New Coordinate Per Query

$$E_0 = \{0\}, \qquad E_t = \text{span}\{e_1, \ldots, e_t\}.$$

For $0 \le t < T$, starting with $\mathcal{K}_0 \subseteq E_0$,

$$\begin{aligned} \mathcal{K}_t \subseteq E_t &\implies y_t \in E_t \\ &\implies \nabla f(y_t) = \frac{L}{4}(A_d y_t - a e_1) \in E_{t+1} \\ &\implies \mathcal{K}_{t+1} \subseteq E_{t+1}. \end{aligned}$$

Example: $d = 7$	1	2	3	4	5	6	7
y_2	$*$	$*$	0	0	0	0	0
$\nabla f(y_2)$	$*$	$*$	$*$	0	0	0	0

$*$: potentially nonzero.

$$\boxed{x_T \in \mathcal{K}_T \subseteq E_T.}$$

Proof: Solve The Chain On A Prefix

For $1 \le m \le d$, let A_m be the leading $m \times m$ block of A_d. The minimizer over E_m obeys

$$A_m u_{1:m} = a(e_1)_{1:m} \quad \Longleftrightarrow \quad 2u_i - u_{i-1} - u_{i+1} = 0 \ (1 \le i \le m),$$

with virtual boundary values $u_0 = a$, $u_{m+1} = 0$.

$$u_i = a\left(1 - \frac{i}{m+1}\right) \ (1 \le i \le m), \qquad u_i = 0 \ (i > m).$$

Restricted Minimum

$$\min_{x \in E_m} f(x) = f(u) = \frac{L}{8}\big(au_1 - 2au_1\big) = -\frac{La^2}{8}\frac{m}{m+1}.$$

Proof: Compare The Two Optimal Values

Restricted And Full Optima: $x_T \in E_T$, $E_d = \mathbb{R}^d$

$$
\begin{aligned}
f(x_T) - f^\star &\geq \min_{x \in E_T} f(x) - \min_{x \in \mathbb{R}^d} f(x) \\
&= \frac{La^2}{8}\left(\frac{d}{d+1} - \frac{T}{T+1}\right) \\
&= \frac{La^2}{8}\left(\frac{1}{T+1} - \frac{1}{d+1}\right) \\
&= \frac{La^2}{16(T+1)}, \qquad d = 2T+1.
\end{aligned}
$$

Proof: Normalize The Initial Distance

For $m = d$, the optimizer is $x_i^\star = a(1 - i/(d+1))$, so

$$\|x^\star\|^2 = \frac{a^2}{(d+1)^2} \sum_{j=1}^{d} j^2 = \frac{a^2 d(2d+1)}{6(d+1)} \le \frac{a^2(d+1)}{3}.$$

Choose

$$a = R\sqrt{\frac{6(d+1)}{d(2d+1)}} \quad \Longrightarrow \quad \|x^\star\| = R, \qquad a^2 \ge \frac{3R^2}{d+1}.$$

Since $d + 1 = 2(T+1)$,

$$\boxed{f(x_T) - f^\star \ge \frac{La^2}{16(T+1)} \ge \frac{3LR^2}{32(T+1)^2}.}$$

Predetermined HB Schedules

Theorem (Ma–Zhang, 2026)

There is an absolute $c > 0$ such that for every $T \geq 2$ and every deterministic predetermined schedule $\eta_t \geq 0$, $\beta_t \in [0, 1)$ independent of f, gradients, and iterates, there exist a convex 1-smooth f and $x^\star \in \arg\min f$ with $\|x_0 - x^\star\| \leq 1$ such that HB initialized by $x_{-1} = x_0$ satisfies

$$f(x_T) - f^\star \geq \frac{c}{T^\varphi \log T}, \qquad \varphi = \frac{1+\sqrt{5}}{2}.$$

Randomized Schedules

Theorem (He–Zhang, 2026)

Let f be convex and L-smooth, $L > 0$, with a minimizer $x^\star$ and $\|x_0 - x^\star\| \le R$. Run their predefined, oracle-independent random HB schedule with exact gradients from $x_{-1} = x_0$, where $\eta_t \ge 0$ and $\beta_t \ge 0$; the momentum β_t may exceed 1. For every deterministic $T \ge 1$,

$$\mathbb{E}[f(x_T) - f^\star] \le \frac{4LR^2}{(1 + T/1024)^{3/2}}.$$

Along the same infinite run,

$$\mathbb{P}\left(\sup_{T \ge 1} T^{3/2}[f(x_T) - f^\star] < \infty\right) = 1.$$

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